
Externalized CPDAG Summaries Improve LLM Causal Deduction

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Abstract

Corr2Cause asks whether a causal claim holds in every DAG compatible with observed correlations and conditional independencies. We frame this as latent-object reasoning: the label is defined by a CPDAG query, but free-form chain-of-thought often collapses the Markov-equivalence-class problem into local pattern matching. We propose *Structured Thinking*, a two-turn pipeline that first externalizes a typed, schema-constrained CPDAG summary and then answers against that graph state. On the Corr2Cause full test, Structured Thinking raises Qwen3.5-27B from 73.0 to 86.4 F_1 (YES) over a strong PC-instruction baseline in the primary paired run (+13.4 pp; McNemar $p = 2.4 \times 10^{-6}$; bootstrap 95% CI [+8.4, +18.6]); across three full-ID seeds, the mean gain is $+8.1 \pm 5.3$ pp. A PC-scaffolded two-turn prose control reaches only 67.6 F_1 , indicating that a detailed PC scaffold plus a schema-free prose intermediate is not sufficient. The same pattern holds on Qwen3.6-27B, Paraphrase-OOD, and GPT-5.4-mini. Scrambling the emitted CPDAG costs 12.0 pp F_1 , and a full-split audit shows close agreement with the reference CPDAG (ID skeleton F_1 0.960; exact match 75.9%). These results support a bounded design principle: externalize the latent object that defines the label, constrain its form, and test whether downstream answers use it.

1 Introduction

Suppose we observe that some variables are correlated, while others become independent after conditioning on a third variable. Under standard causal assumptions, these statistical facts restrict which causal graphs could have generated the data. They usually do not identify a single graph. Instead, they define a family of compatible graphs.

This creates a global deduction problem. Some causal claims are *forced*: they hold in every compatible graph. Others are merely *possible*: they hold in some compatible graphs but fail in others. A model

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that finds one plausible causal story has not solved the task; it must check whether the claim survives all valid graph completions.

Corr2Cause [Jin et al., 2024] turns this problem into a benchmark for language models. Each instance gives marginal dependencies and conditional independencies, then asks whether a natural-language causal hypothesis is deducible. The hidden object behind the label is the compatible graph family. In causal discovery, this family is compactly represented by a completed partially directed acyclic graph, or CPDAG: directed edges are forced across the whole Markov equivalence class, while undirected edges encode genuine ambiguity.

This makes Corr2Cause a latent-object reasoning task. To answer correctly, a model must keep track of a mixed graph object: which variable pairs are adjacent, which non-adjacent pairs have which separating sets, which triples form compelled colliders, which orientations are forced, and which edges remain ambiguous. Free-form chain-of-thought is a weak state representation for this object. It often produces locally plausible reasoning while collapsing the global “all compatible graphs” query into pattern matching over a few adjacencies or partially oriented edges.

We propose *Structured Thinking*, a two-turn prompting-and-decoding pipeline designed around this latent object. In Turn 1, the model must emit a typed CPDAG-style summary with fields for variables, skeleton edges, non-adjacent pairs and separating sets, v-structures, directed edges, and undirected edges. For Qwen-family open-weight models, decoding is constrained to this schema; for API models, the same object is enforced through tool calling. In Turn 2, the emitted graph summary is returned to the model, which answers the causal hypothesis against the materialized state. The method does not fine-tune the model and does not call a symbolic causal solver. Its purpose is narrower: replace an unstable prose trace with a constrained graph-shaped anchor for the final decision.

Our main direct-answer comparison is a strong PC-instruction baseline rather than plain chain-of-thought. The one-turn baseline is told to remove edges using conditional independencies, orient v-structures, propagate forced orientations, and answer YES only when the hypothesis holds in every compatible DAG. This controls for explicit PC-algorithm content, although the structured prompt is not token-identical and the final method still bundles externalization, schema constraints, and Turn 2 graph-query instructions. We also add a PC-scaffolded two-turn prose control: Turn 1 uses the same causal-discovery scaffold as the direct baseline but writes free text, and Turn 2 answers from that analysis. The prose control tests whether the same PC scaffold plus a schema-free intermediate is sufficient without requiring a typed CPDAG object.

Empirically, Structured Thinking substantially improves causal deduction on Corr2Cause. On the full test, the primary Qwen3.5-72B run improves from 73.0 to 86.4 F_1 (YES), a +13.4 pp gain over the PC-instruction baseline, and two additional full-test seeds keep the gain positive (+8.1 $\pm$ 5.3 pp across three seeds). The PC-scaffolded prose control reaches only 67.6 F_1 , so a detailed PC scaffold plus a schema-free intermediate is not sufficient. The same pattern holds on Qwen3.6-72B (+9.8 pp), persists on a Paraphrase-OOD split (+10.5 pp), and remains positive for GPT-5.4-mini (+3.2 pp). Finally, graph-content probes and a full-split CPDAG-field audit show that the intermediate is not decorative: scrambling the emitted CPDAG summary drops Qwen3.5-72B by 12.0 pp F_1 , the emitted graph exactly matches the reference CPDAG in about three quarters of cases, and final-answer accuracy degrades monotonically with graph quality.

Contributions.

- We frame Corr2Cause as a *latent-object reasoning* problem: the label is defined by a CPDAG query, so the model should be evaluated on whether it can construct and use that graph object.
- We introduce a two-turn, schema-constrained CPDAG-summary pipeline that externalizes this object before answering, without fine-tuning or a symbolic causal solver.
- We provide evidence against strong direct-answer PC baselines across open-weight and API models, together with Paraphrase-OOD evaluation, ablations, graph-content probes, and a full-split CPDAG-field audit showing that the emitted graph state is usually close to the reference object and predictive of downstream correctness.

2 Background and Related Work

CPDAGs and causal deduction. The PC algorithm [Kalisch and Bühlmann, 2007, Spirtes et al., 2001] removes edges using conditional independencies, orients v-structures, and propagates Meek rules [Meek, 1995]. Its output is a CPDAG: directed edges are forced across the equivalence class, while undirected edges encode genuine ambiguity [Pearl, 2009]. Corr2Cause uses this CPDAG as the hidden object behind each label. This is why the task is a better stress test than asking whether two variables are merely associated: an answer can be wrong even when the model finds a plausible DAG, if another DAG in the same equivalence class falsifies the hypothesis.

LLMs and structural causality. Surveys and benchmarks consistently find structural causal reasoning harder for LLMs than commonsense causal association [Kiciman et al., 2024, Ma, 2025, Zečević et al., 2023, Willig et al., 2022, Long et al., 2023, Chi et al., 2024]. Yang et al. [2024] argue that many causal benchmarks can be partially solved by retrieving familiar domain knowledge; Corr2Cause is useful precisely because its labels are generated from independence constraints and PC-algorithm semantics. Complementary benchmarks such as CLEAR-3K and CausalARC evaluate natural-language causal explanation and SCM-world reasoning rather than CPDAG deduction [Liu et al., 2026, Maasch et al., 2025]; this motivates the scope boundary in §8. Prompting strategies improve performance [Sgouritsa et al., 2024, Liu et al., 2025, Bagheri et al., 2024], but still leave a gap to algorithmic ground truth, especially as the number of variables grows.

Structured intermediates. The closest precursor to this work builds a knowledge-graph style intermediate for Corr2Cause and shows that explicit structure can improve causal generalization [Sun et al., 2025]. We make the latent object more specific: the intermediate is a CPDAG summary, emitted under regex- or JSON-schema-constrained decoding [Willard and Louf, 2023, Dong et al., 2024], and compared against a strong direct-answer PC baseline. Concurrent modular prompting decomposes Corr2Cause into prose sub-answers [Kadziolka and Salehkaleybar, 2025]; C2P similarly uses causal prompting steps but is not a constrained CPDAG-state method [Bagheri et al., 2024]. A prose sub-answer can be locally plausible but hard to audit; a typed graph has explicit fields whose content can be removed, scrambled, or queried.

Faithfulness of reasoning traces. Recent work questions whether free-form CoT is causally used by the final answer, using pre-CoT probes, causal-bypass diagnostics, causal sufficiency/necessity tests, and judge benchmarks for process faithfulness [Cox et al., 2026, Sathyanarayanan et al., 2026, Yu et al., 2025, Mittal and Arike, 2026, Fu et al., 2025]. Our probes are behavioral rather than activation-level: after Turn 1 emits a graph, we intervene on graph fields and measure Turn 2 degradation. They therefore test content dependence of the emitted state, not full mechanistic faithfulness of every token.

Tool use, programs, and positioning. Tool-augmented language models [Parisi et al., 2022, Yao et al., 2023, Schick et al., 2023] usually call external functions, while program-aided prompting [Gao et al., 2023, Chen et al., 2023] offloads computation to code. Structured Thinking instead uses the tool interface to record the model’s own schema-constrained analysis; no symbolic causal solver supplies the answer. The claim is intentionally narrow: for a CPDAG-query task, a constrained CPDAG-summary intermediate beats a strong PC-scaffolded direct baseline and a PC-scaffolded two-turn prose control. The broader recipe is to identify the latent object that defines the label, externalize it, and test whether downstream answers depend on its content.

3 Method

3.1 Task

An instance has a premise P listing marginal dependencies and conditional independencies $X_i \perp X_j \mid Z$, plus a hypothesis H such as “ X_a is a parent of X_b .” Under the faithfulness and Markov assumptions, P identifies a CPDAG $\mathcal{G}$ and its Markov equivalence class $\text{MEC}(\mathcal{G})$. The label is

$$y(P, H) = \text{YES} \iff \forall D \in \text{MEC}(\mathcal{G}) : H \text{ holds in } D,$$

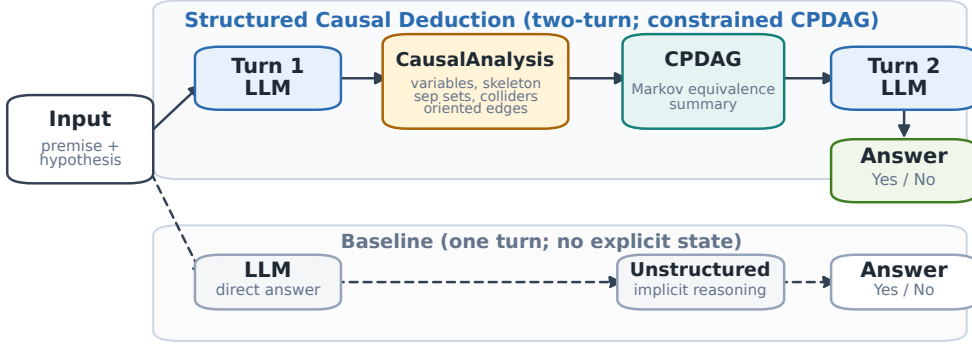


Figure 1: **Structured Causal Deduction Pipeline.** Turn 1 emits a schema-constrained CPDAG summary; Turn 2 answers the hypothesis against that graph. The baseline receives explicit PC-algorithm guidance but answers directly, so the comparison tests structured externalization against a strong direct-answer PC baseline.

i.e. the hypothesis must survive *all* valid DAG completions. We report F_1 on the positive class as the primary metric because the ID split is imbalanced toward NO (180/1162 positives), and report accuracy as a secondary metric.

3.2 Structured Thinking pipeline

Structured Thinking is a two-turn conversation. In Turn 1, the system prompt describes the PC algorithm and requires a `CausalAnalysis` object with fields `variables`, `skeleton`, `non_adjacent`, `v_structures`, `directed`, and `undirected`. For Qwen-family open-weight models, decoding is constrained by a schema-derived regex; for API models, the same schema is enforced through JSON-schema tool calling. Gemma uses the same structured interface without hard constrained decoding because the available backend was unreliable for that tokenizer. Thinking tokens, when used, appear before the constrained region. In Turn 2, the emitted object g is returned as a tool-result message with a compact DOT view, and the model answers H by checking adjacency, forced orientations, and directed reachability over all valid completions. The final YES/NO is extracted by regex.

Throughout the paper, “CPDAG” denotes the intended semantics of this emitted summary. The schema enforces parseable fields, not acyclicity, Meek closure, or full CPDAG consistency; schema-valid but graph-inconsistent cases are audited explicitly in §7.1. The design pins both the intermediate *type* (a CPDAG summary rather than prose) and the Turn 2 *target* (a graph query rather than a restatement of P). We ablate the PC prompt, constrained decoding, thinking tokens, premise visibility, and graph content in §5–§6; full prompt and schema text appears in Appendix E–F.

Why this intermediate is useful. A CPDAG separates three decisions that free text tends to entangle: which pairs are adjacent, which triples are compelled colliders, and which orientations are forced across the equivalence class. Turn 2 can inspect this materialized structure rather than rediscover it from the premise. The summary is also compact: a six-variable instance becomes short edge and v-structure lists rather than a long derivation. The intended advantage over unconstrained chain-of-thought is not more verbosity, but a better state representation for the specific decision.

Constrained decoding. The constraint is applied to the *analysis object*, not to the free reasoning tokens that precede it. For Qwen-family models, the regex wraps a `<think>` block followed by a `<tool_call>` envelope; for API models, the same schema is enforced through tool calling. Gemma runs keep the same structured interface but omit hard regex constraints. Removing the hard constraint keeps the tool-call template as a soft instruction, so the ablation asks whether syntactic enforcement itself matters beyond reminding the model to be structured.

3.3 Baselines and controls

The principal baseline, `BASELINEENHANCED`, is a strong direct-answer PC baseline. It receives an explicit PC-algorithm walk-through in a single turn and answers directly. The prompt covers skeleton removal from conditional independencies, collider orientation, Meek-rule propagation, closed-world treatment of unstated independencies, and the requirement that a positive hypothesis hold in every DAG in the Markov equivalence class. This comparison controls for giving the direct-answer model an explicit PC-algorithm scaffold, although the structured prompt is not token-identical and the final method still bundles externalization, schema constraints, and Turn 2 graph-query instructions.

As a prose-state control, we add `TWOTURN-PROSE-PC`. Turn 1 uses the same detailed PC scaffold as `BASELINEENHANCED` but asks for free-text analysis while withholding the final yes/no answer; Turn 2 returns that analysis and requests the answer. This control uses no tool call and no schema-constrained typed graph object. Its role is narrow: it tests whether the same PC scaffold plus a two-turn prose intermediate are sufficient, while `BASELINEENHANCED` remains the one-turn PC-instruction control.

We also include a single-turn structured-prompt diagnostic for Qwen3.6-27B, `SINGLETURN-STRUCTPROMPT`. This condition reuses the structured CPDAG-oriented instructions but does not require a tool call or schema-constrained graph output before answering. It tests whether merely mentioning the right graph vocabulary can replace explicit graph-object externalization.

Structured Thinking is the only condition that materializes a typed graph summary before the final decision. The probes in §6 then test whether Turn 2 actually depends on the graph content.

4 Main Results

4.1 Setup

Unless noted, open-weight models (Qwen3.5-27B, Qwen3.6-27B, Qwen3.5-9B, Gemma-4-31B-it) run under vLLM [Kwon et al., 2023] with $T=0.6$, $\text{top-}p=0.95$, $\text{max_model_len}=32768$, and fixed seed; gpt-5.4-mini uses the OpenAI Responses API with `reasoning_effort="high"`. Tables use short names; the public model IDs are Qwen/Qwen3.5-27B, Qwen/Qwen3.6-27B, google/gemma-4-31B-it, and gpt-5.4-mini, with the OpenAI snapshot and Hugging Face revision hashes recorded in per-experiment metadata. ID is the $n=1162$ Corr2Cause test split; Paraphrase-OOD is a surface-refactored split with $n=2246$ that preserves graph instances while changing wording. Qwen-family Structured Thinking runs use schema-derived regex constraints via xgrammar [Dong et al., 2024]; direct baselines and `TWOTURN-PROSE-PC` do not emit a constrained CPDAG object. Gemma Structured is reported without constrained decoding because available backends were unreliable for that tokenizer. For the principal Qwen3.5-27B ID comparison we report a $B=10,000$ paired-bootstrap CI and an exact McNemar test on the primary seed, and Appendix B reports two additional full-ID reruns on local seeds 52 and 62. Other cross-model and Paraphrase-OOD rows are fixed-seed unless noted.

4.2 Cross-model results

Table 1 gives the primary paired comparison. On Qwen3.5-27B ID, Structured Thinking improves $F_1(\text{YES})$ from 73.01 to 86.36, a +13.35 pp gain with bootstrap CI [+8.42, +18.59] and exact McNemar $p = 2.4 \times 10^{-6}$. In this primary seed, the gain is not a precision-recall tradeoff: true positives rise 119→152, false negatives fall 61→28, and false positives fall 27→20 (Appendix Table 5).

We further reran the full ID split on two additional local seeds (52 and 62). Structured Thinking remains ahead on both reruns, with F_1 gains of +2.68 pp and +8.21 pp and accuracy gains of +0.60 pp and +1.89 pp, respectively. Across the three full-ID seeds, it averages $85.57 \pm 1.70 F_1$ versus 77.49 ± 4.07 for `BASELINEENHANCED` ($\Delta = +8.08 \pm 5.34$ pp). The error shift is seed-dependent: the additional seeds trade small precision decreases for larger recall improvements, so the robust pattern is improved positive-class F_1 and recall rather than uniformly higher precision.

The PC-scaffolded prose control sharpens the interpretation. `TWOTURN-PROSE-PC` reaches only 67.55 F_1 , below both the one-turn PC-instruction baseline and Structured Thinking. It has higher recall than the one-turn baseline, but at the cost of many more false positives (71 versus 27), showing

Table 1: **Main results on Corr2Cause.** $F_1(\text{YES})$ and accuracy (%). Δ is the absolute F_1 change relative to the one-turn PC-instruction baseline for the same model and split. The principal Qwen3.5-27B ID comparison has McNemar $p = 2.4 \times 10^{-6}$ and bootstrap 95% CI [+8.42, +18.59].

Model	Method	Split	$F_1(\text{Yes})$	Acc.	ΔF_1
Qwen3.5-27B	BaselineEnhanced	ID	73.01	92.43	—
	TwoTurn-Prose-PC	ID	67.55	89.41	−5.46
	Structured Thinking	ID	86.36	95.87	+13.35
	BaselineEnhanced	Paraphrase-OOD	72.33	92.61	—
	Structured Thinking	Paraphrase-OOD	82.79	95.06	+10.46
Qwen3.6-27B	BaselineEnhanced	ID	75.92	92.69	—
	SingleTurn-StructPrompt	ID	74.53	91.82	−1.39
	Structured Thinking	ID	85.71	95.52	+9.79
Gemma-4-31B	BaselineEnhanced	ID	85.38	95.70	—
	Structured Thinking [†]	ID	87.39	96.13	+2.01
Qwen3.5-9B	BaselineEnhanced	ID	39.18	84.77	—
	Structured Thinking	ID	59.56	88.90	+20.38
GPT-5.4-mini	BaselineEnhanced	ID	85.39	95.61	—
	Structured Thinking	ID	88.58	96.47	+3.19

[†] Gemma Structured is reported without hard constrained decoding because the available backend was unreliable for that tokenizer.

that a schema-free prose intermediate under the same PC scaffold mainly changes answer bias rather than producing a reliable graph state in this run. Structured Thinking instead reduces both false negatives and false positives on the primary seed.

The same pattern holds beyond one Qwen release. On Qwen3.6-27B, Structured Thinking improves from 75.92 to 85.71 F_1 (+9.79 pp). A single-turn structured-prompt diagnostic reaches 74.53, so merely giving CPDAG-oriented wording without requiring an external graph object does not explain the gain.

The cross-family rows are supportive but should be read with backend asymmetries in mind. Qwen3.5-27B + Structured Thinking (86.36) matches or exceeds the direct PC baselines of Gemma-4-31B and GPT-5.4-mini (both about 85.4), but remains below their structured runs. The result therefore narrows much of the gap to stronger closed/API or larger-model baselines on this benchmark, rather than eliminating it. Gemma gains +2.01 pp despite lacking hard constrained decoding, while GPT-5.4-mini gains +3.19 pp under the same structured interface. On the Paraphrase-OOD split, Qwen3.5-27B retains a +10.46 pp advantage, suggesting that the intervention is not tied to the original Corr2Cause surface templates.

4.3 Effect scales with task complexity

Figure 2 shows that the gain is larger on the two hardest informative bins, exactly where MEC reasoning is hardest: ID improves by +5.3 pp at $n=5$ and +18.6 pp at $n=6$; Paraphrase-OOD improves by +5.9 and +13.6 pp. Lower-variable bins have too few positives for stable F_1 estimates, so we avoid treating this as a full trend analysis. The pattern supports the interpretation that the CPDAG intermediate helps most when local free-text heuristics break down.

Appendix C extends this check with pilot larger-graph stress tests. These rows are not a controlled variable-count sweep because the graph families differ, but they test whether the qualitative advantage survives beyond the original $n \leq 6$ regime. On Qwen3.5-27B, Structured Thinking improves Random $n=7$ from 28.42 to 91.87 F_1 , ALARM-derived $n=7$ from 15.38 to 88.73, and ASIA-derived $n=8$ from 9.30 to 60.00. For named-network-derived rows, source names and semantic variable names are withheld from the model. Qwen3.5-9B also improves in every setting but with much lower absolute F_1 , suggesting that larger-graph CPDAG construction remains capacity-sensitive.

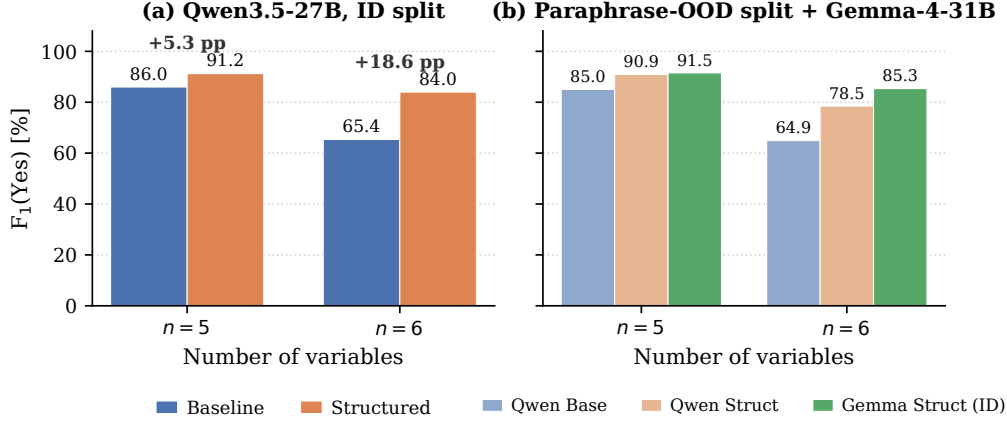


Figure 2: **F₁ by variable count.** The Qwen3.5-27B advantage widens from +5.3 pp at $n=5$ to +18.6 pp at $n=6$ on ID, and from +5.9 to +13.6 pp on Paraphrase-OOD. Bars at $n \leq 4$ have too few positives to be informative and are omitted. Gemma is shown only as an ID reference point, not as part of the Paraphrase-OOD comparison.

Table 2: **Ablations on Qwen3.5-27B.** $n=1162$ ID split. Removing the PC-algorithm prompt collapses the method almost entirely; removing thinking or constrained decoding each causes a substantial drop.

Configuration	F ₁ (Yes)	Acc.	Δ vs. full
Structured Thinking (full)	86.36	95.87	—
–thinking	71.95	92.08	–14.41
–constrained decoding	76.88	93.37	–9.48
–PC-algorithm prompt	36.21	87.26	–50.15
BaselineEnhanced	73.01	92.43	–13.35

5 Ablations

We stress-test the Qwen3.5-27B gain by removing one design choice at a time (Table 2). These one-factor removals are not a full factorial decomposition, so the numbers should be read as sensitivity tests rather than additive effect estimates.

The PC-algorithm prompt is the dominant contributor: replacing it with a minimal “produce a causal analysis” instruction drops F₁ to 36.2, far below baseline. This is useful evidence in our favor because it rules out the shallow interpretation that “any tool call” solves the benchmark. The causal-discovery content must be present.

On top of that prompt, removing constrained decoding costs 9.5 pp and removing thinking costs 14.4 pp at 27 B. The constrained-decoding ablation is especially informative: the model still sees tool-call syntax, but the hard regular-language constraint is removed. The drop therefore indicates that the constraint is not just preventing parse errors; it regularizes the generation toward a coherent CPDAG-shaped state. The full-split probes below test graph content directly, rather than relying on small field-drop subsets.

A smaller-model factorial is reported in Appendix D: at 9 B, thinking becomes harmful while constrained decoding remains helpful, suggesting that explicit reasoning traces should be calibrated by model size.

6 Probes: is the emitted graph actually used?

If the gain came only from extra Turn 1 tokens or tool-call formatting, corrupting the CPDAG content in Turn 2 should have little effect. We test this on the full Qwen3.5-27B ID split (Table 3).

Table 3: **Turn-2 probes on Qwen3.5-27B ID**. Full Structured is the reference. Scrambling the full CPDAG costs 12.0 pp, and corrupting individual components each costs about 6 pp.

Turn-2 condition	F ₁ (Yes)	Acc.	ΔF_1
Full Structured (reference)	86.36	95.87	—
no-premise (drop P in Turn 2)	82.56	94.84	−3.80
scrambled-skeleton only	80.11	93.98	−6.25
scrambled-v-structures only	80.00	94.06	−6.36
scrambled-directed only	80.70	94.32	−5.66
full-scramble (all of skeleton, directed, undirected)	74.32	92.69	−12.04

Table 4: **Automatic CPDAG-field audit** for Structured Thinking on full ID and Paraphrase-OOD splits. Graph-consistency violations are broad graph-semantic failures among schema-valid outputs, including field conflicts, v-structure inconsistencies, missing closure, or no valid DAG extension.

Audit metric	ID	Paraphrase-OOD
Skeleton F ₁	0.960	0.959
Directed-edge F ₁	0.896	0.884
V-structure F ₁	0.923	0.920
Exact CPDAG match	75.9%	74.4%
Graph-consistency violations	18.4%	20.7%
Acc. when graph exact	97.2%	96.6%
Acc. with consistency violation	91.1%	90.1%

The full scramble drops F₁ by 12.0 pp, far more than removing the premise from Turn 2 (3.8 pp); the pipeline is therefore not just re-reading the premise. Skeleton, v-structure, and directed-edge scrambles each lower F₁ by 5.7–6.4 pp, ruling out both a pure adjacency-extraction story and a shallow answer-key story. These are destructive behavioral probes: they demonstrate dependence on graph content, not proof that Turn 2 performs a formally valid MEC proof on every instance, because a scrambled graph may no longer be a valid CPDAG.

7 Analysis and Practitioner Findings

7.1 Error localization and error shifts

Structured Thinking makes failures easier to diagnose because the intermediate graph can be inspected separately from the final answer. At the label level, the primary-seed change is clear from Appendix Table 5: Qwen3.5-27B reduces positive-class misses from 61 to 28 and false positives from 27 to 20. Thus the method does not merely make the model more willing to answer YES in that run; it improves recall while also slightly improving specificity.

Automatic CPDAG-field audit. We next compare the emitted `CausalAnalysis` object against the reference CPDAG fields on the full ID and Paraphrase-OOD splits. This audit separates final-answer accuracy from graph construction quality. On ID, Structured Thinking achieves 0.960 skeleton F₁, 0.923 v-structure F₁, 0.896 directed-edge F₁, and 75.9% exact CPDAG match; the corresponding Paraphrase-OOD values are similar (0.959, 0.920, 0.884, and 74.4%). Thus the intermediate is usually not merely graph-shaped text: its fields agree closely with the latent CPDAG that defines the label.

At the same time, the audit quantifies the remaining gap between schema validity and graph-semantic validity. Graph-consistency violations occur in 18.4% of ID outputs and 20.7% of Paraphrase-OOD outputs, with Meek-closure violations in 5.1% and 6.0%, respectively. These failures are not uniformly catastrophic: many are local or irrelevant to the queried hypothesis, and Turn 2 still sees the premise and hypothesis. Nevertheless, graph quality is predictive of downstream correctness. ID accuracy drops from 97.2% when the emitted CPDAG exactly matches the reference to 93.9% for valid but non-exact graphs and 91.1% for summaries with graph-consistency violations; Paraphrase-OOD shows the same ordering (96.6%, 92.8%, 90.1%).

A 32-case manual diagnostic in Appendix I gives the same qualitative picture: many residual errors are already visible in the externalized graph state, while others come from Turn-2 readout or MEC-ambiguity handling.

7.2 Ruling out simpler explanations

The main alternative explanation is that Structured Thinking is merely a stronger prompt. BASELINEENHANCED makes that reading less plausible rather than impossible: the direct-answer model receives an explicit PC-algorithm scaffold and the same user instance, so the +13.35 pp gain is not a plain-CoT comparison. The Qwen3.6 single-turn structured-prompt diagnostic further supports this point: reusing the CPDAG-oriented wording without forcing an external graph object reaches 74.53 F_1 , below the standard Qwen3.6 baseline and far below Structured Thinking’s 85.71.

A second alternative is that the gain comes from the extra interaction rather than from the typed graph state. The TWOTURN-PROSE-PC control addresses this by using the same PC scaffold in a two-turn prose workflow without a schema-constrained graph object. It reaches 67.55 F_1 on Qwen3.5-27B, below the one-turn PC baseline (73.01) and far below Structured Thinking (86.36). Thus, a detailed PC scaffold plus a schema-free intermediate is not sufficient. The corresponding claim is narrow: the observed gain is not explained by the two-turn prose interface alone.

A third alternative is superficial tool formatting. The probes in §6 argue against this: dropping the premise in Turn 2 costs only 3.8 pp, while corrupting the graph costs 12.0 pp, and separate skeleton, v-structure, and directed-edge scrambles each cost about 6 pp. The CPDAG-field audit in Table 4 adds the constructive counterpart: the naturally emitted graph fields closely match the reference object in most cases, and downstream accuracy falls as graph quality falls. The downstream answer therefore depends on the content of the emitted graph, not just on the existence of a tool call.

7.3 Cost, reproducibility, and deployment

Structured Thinking requires two model calls and, for Qwen-family local runs, a schema-constrained decoder, but no fine-tuning, learned verifier, or symbolic causal solver at inference time. The token footprint is roughly $2\times$ the one-turn baseline and wall-clock is about $1.5\times$ in batched vLLM runs. API-backed reasoning models require care: strict tool forcing can compress reasoning before the tool call, while `tool_choice="auto"` restored GPT-5.4-mini performance in our runs (Appendix A.1). The strongest result is reproducible with an open model (Qwen3.5-27B); code, data, prompts, schemas, and metadata are released (Appendix J). The repository includes per-run evaluation, usage, and metadata files, so appendix rows can be audited without rerunning the models. In practice, the components should be calibrated on a validation split: verify the PC prompt, test constrained decoding, and enable thinking only when traces are reliable at the target scale.

8 Limitations

Our claims are scoped to CPDAG-style deduction from conditional independence statements. The same design principle may transfer to SCM queries, intervention benchmarks, or world-model settings, but the intermediate representation would need to change and this paper does not establish that transfer.

The principal Qwen3.5-27B ID comparison is supported by three full-test local seeds, with F_1 and accuracy improving on each. However, Paraphrase-OOD, larger-graph stress tests, cross-model rows, and several controls remain single-seed because of cost, so smaller gains on already strong baselines should be read directionally.

The emitted `CausalAnalysis` is schema-valid but not certified: the schema does not enforce acyclicity, Meek closure, or full internal consistency. The audit in Table 4 quantifies this gap: graph-consistency violations occur in 18–21% of outputs, and certified CPDAG validators remain useful future work.

Finally, the controls do not fully decompose every component of Structured Thinking. They control for a strong one-turn PC-instruction baseline, a PC-scaffolded two-turn prose interface without typed schema, and superficial tool formatting, but schema constraints, graph-field names, tool-result formatting, and Turn-2 query instructions are bundled in the final method. Lightly structured prose

intermediates and validity-preserving graph interventions would further sharpen the mechanism claim.

9 Conclusion

Structured Thinking turns causal deduction from free-form prose into a typed CPDAG-summary query. Against a strong direct-answer PC baseline, it yields a statistically significant +13.35 pp F_1 gain on Qwen3.5-27B in the primary paired run and remains positive across two additional full-ID seeds; it also persists on Paraphrase-OOD, holds on Qwen3.6-27B, and remains positive on GPT-5.4-mini. A PC-scaffolded two-turn prose control and a single-turn structured-prompt diagnostic do not recover the gain, while ablations, graph scrambles, and the CPDAG-field audit show that the PC prompt, decoding constraint, and graph content are load-bearing. The bounded lesson is to externalize the latent formal object that defines the label, constrain its form, and test whether downstream reasoning uses it.

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A Additional diagnostics

Table 5: **Confusion matrices, Qwen3.5-27B ID split ($n=1162$).** Structured Thinking improves both false negatives ($61 \rightarrow 28$) and false positives ($27 \rightarrow 20$) on the primary seed. The PC-scaffolded prose control increases recall but introduces many false positives.

Method	Actual	Predicted		TP	FN	FP
		NO	YES			
BaselineEnhanced	NO	955	27	119	61	27
	YES	61	119			
TwoTurn-Prose-PC	NO	911	71	128	52	71
	YES	52	128			
Structured Thinking	NO	962	20	152	28	20
	YES	28	152			

A.1 API configuration sensitivity

On API-backed reasoning models, strict `tool_choice="required"` can change the reasoning regime. In our initial GPT-5.4-mini run, name-forcing the `CausalAnalysis` tool reduced F_1 to 74.7 (-10.7 pp vs. the 85.4 baseline): median reasoning tokens on $n=6$ fell from roughly 5000 to roughly 800, and 54% of emitted CPDAGs had zero directed edges. Switching to `tool_choice="auto"` plus a prompt-level instruction to call the tool preserved the same schema while restoring the reasoning budget (roughly 5600 tokens on $n=6$), raising the directed-edge rate from 42% to 96% and recovering the reported 88.6 F_1 . The two configurations differ only in this API flag and prompt-level instruction.

B Full-ID seed robustness

Because the local Qwen runs use stochastic decoding, we rerun the principal Qwen3.5-27B ID comparison on two additional full-test seeds. Table 6 reports the full ID split ($n=1162$). Structured Thinking improves F_1 and accuracy on every full run, although the size of the gain varies across seeds.

The primary seed improves both precision and recall, while the two additional seeds are mainly recall-driven and incur small precision costs. We therefore treat the multi-seed pattern as evidence for a positive F_1 /accuracy effect on the full ID split, not as evidence that every seed improves every error type.

Table 6: **Full-ID seed robustness for Qwen3.5-27B on Corr2Cause** ($n=1162$). Values are percentages. Seed 42 is the primary run from Table 1; seeds 52 and 62 are additional full-split reruns.

Seed	Method	F ₁ (Yes)	Acc.	Precision	Recall
42	BaselineEnhanced	73.01	92.43	81.51	66.11
42	Structured Thinking	86.36	95.87	88.37	84.44
52	BaselineEnhanced	80.94	94.41	85.71	76.67
52	Structured Thinking	83.62	95.01	85.06	82.22
62	BaselineEnhanced	78.53	93.98	87.67	71.11
62	Structured Thinking	86.74	95.87	86.26	87.22
BaselineEnhanced, mean $\pm$ std		77.49 $\pm$ 4.07	93.61 $\pm$ 1.04	–	–
Structured Thinking, mean $\pm$ std		85.57 $\pm$ 1.70	95.58 $\pm$ 0.50	–	–
Δ Structured – Baseline		+8.08 $\pm$ 5.34	+1.98 $\pm$ 1.42	–	–

C Larger-graph stress tests

Table 7: **Pilot larger-graph stress tests**. Values are percentages except N and #Yes. These single-seed rows use graph families outside the original Corr2Cause $n \leq 6$ range and are intended as stress tests rather than replacement main results.

Setting	Model	Method	N	#Yes	Prec.	Rec.	Acc.	F ₁
Random $n=7$ pilot	Qwen3.5-27B	Baseline	300	150	67.50	18.00	54.67	28.42
Random $n=7$ pilot	Qwen3.5-27B	Structured	300	150	97.74	86.67	92.33	91.87
Random $n=7$ pilot	Qwen3.5-9B	Baseline	300	150	75.76	16.67	55.67	27.32
Random $n=7$ pilot	Qwen3.5-9B	Structured	300	150	70.37	25.33	57.33	37.25
ALARM-derived $n=7$	Qwen3.5-27B	Baseline	300	150	43.75	9.33	48.67	15.38
ALARM-derived $n=7$	Qwen3.5-27B	Structured	300	150	97.60	81.33	89.67	88.73
ALARM-derived $n=7$	Qwen3.5-9B	Baseline	300	150	47.83	14.67	49.33	22.45
ALARM-derived $n=7$	Qwen3.5-9B	Structured	300	150	65.38	22.67	55.33	33.66
ASIA-derived $n=8$	Qwen3.5-27B	Baseline	168	20	8.70	10.00	76.79	9.30
ASIA-derived $n=8$	Qwen3.5-27B	Structured	168	20	60.00	60.00	90.48	60.00
ASIA-derived $n=8$	Qwen3.5-9B	Baseline	40	20	100.00	5.00	52.50	9.52
ASIA-derived $n=8$	Qwen3.5-9B	Structured	40	20	100.00	20.00	60.00	33.33

The 27B gains are large on all three stress settings, while the 9B rows remain positive but capacity-limited. The ASIA-derived 9B row is especially small ($N=40$) and should be read directionally. The source graph families are used to generate larger CPDAG-query instances; source-network names and semantic variable names are withheld from the model, and prompts use anonymized variable labels. These rows are not meant to replace the paired full-ID comparison in Table 1. Seed metadata and confusion matrices are included in the released evaluation files.

D Scale interaction diagnostic

Table 8: **Qwen3.5-9B factorial** ($n=1162$, ID split). Thinking is net-harmful to Structured Thinking at 9 B, reversing its Qwen3.5-27B behaviour (Table 2).

Configuration	F ₁ (Yes)	Acc.
Structured Thinking (full)	59.56	88.90
–thinking	64.46	89.85
–constrained decoding	45.85	85.97
BaselineEnhanced	39.18	84.77
–thinking	45.58	86.23

Thinking helps at 27 B but hurts at 9 B: disabling it improves Structured Thinking by +4.9 pp and the baseline by +6.4 pp. Constrained decoding shows the opposite pattern, helping more at 9 B (13.7 pp) than at 27 B (9.5 pp). This suggests a practical calibration rule: enable thinking only when a validation split shows that the model’s trace is reliable at the target scale.

E Prompt templates

We reproduce the main prompts for Structured Thinking and the direct PC baseline, as found in `src/prompts.py`. Each prompt is a Python string constant; the Python-level newlines below correspond exactly to the newlines passed to the model. Additional control variants, including the run reported here as `TWO TURN-PROSE-PC` and `SINGLE TURN-STRUCTPROMPT`, are recorded in the released experiment metadata.

E.1 Structured Thinking system prompt (Sstruct, Turn 1)

Source: `src/prompts.py:131-162` (constant `SYSTEM_STRUCTURED_V2`).

You are an expert in causal inference and the PC algorithm for causal discovery from observational data.

Given correlational premises (marginal dependencies and conditional independencies) and a causal hypothesis, analyze the causal structure using the PC algorithm:

```
## Step 1: Determine the Skeleton
- Start with a complete undirected graph over all variables.
- For each conditional independence  $X \perp\!\!\!\perp Y \mid S$ , REMOVE edge  $X$ - $Y$ .
  Record  $S$  as the separating set  $\text{sep}(X,Y)$ .
- CRITICAL: If two variables are adjacent in a faithful DAG, they
  CANNOT be conditionally independent given ANY subset. So any stated
  CI  $X \perp\!\!\!\perp Y \mid S$  proves there is NO direct edge  $X$ - $Y$ .
- Marginal correlation does NOT imply adjacency -- variables can
  correlate through indirect paths.

## Step 2: Identify V-Structures (Colliders)
- For each non-adjacent pair  $(X, Y)$  with common neighbor  $Z$  in
  the skeleton:
  - If  $Z$  is NOT in  $\text{sep}(X,Y)$  -> orient as  $X \rightarrow Z \leftarrow Y$  (v-structure)
  - If  $Z$  IS in  $\text{sep}(X,Y)$  -> NOT a v-structure, do not orient

## Step 3: Apply Meek's Rules (iterate until no changes)
- R1:  $X \rightarrow Y \dashrightarrow Z$ ,  $X$  and  $Z$  non-adjacent => orient  $Y \rightarrow Z$ 
- R2:  $X \rightarrow Y \rightarrow Z$ ,  $X \dashrightarrow Z$  => orient  $X \rightarrow Z$ 
- R3:  $X \dashrightarrow Z$ ,  $Y \rightarrow Z$ ,  $W \rightarrow Z$ ,  $X$  adj  $Y$ ,  $X$  adj  $W$ ,  $Y$  and  $W$ 
  non-adjacent => orient  $X \rightarrow Z$ 

## Step 4: Record your CPDAG using the CausalAnalysis tool
```

Important principles:

- Closed-world assumption: only the stated independencies hold.
Any independence not listed is assumed to NOT hold.
- A hypothesis is deducible iff it holds in ALL DAGs of the
Markov equivalence class (all valid completions of the CPDAG).

End your final answer with exactly "Answer: Yes" or "Answer: No".

Although this legacy Turn-1 system prompt contains an answer-format sentence, the constrained decoder accepts only the `CausalAnalysis` tool call in Turn 1; final answers are generated and extracted only in Turn 2.

E.2 Turn-2 tool-result template

Source: `src/prompts.py:166-187` (function `make_tool_result_v2`). After Turn 1 emits a `CausalAnalysis` object g , the tool-result message fed back in Turn 2 is formatted as:

```
Causal analysis recorded. Your CPDAG summary:
Variables: <variables>
Skeleton: <skeleton>
Non-adjacent: <non_adjacent>
V-structures: <v_structures>
```

Directed: <directed>
Undirected: <undirected>

<DOT visualization>

Now evaluate the hypothesis against this CPDAG:

1. Check if the relevant edge exists in the skeleton.
2. Check if its direction is determined in the directed edges.
3. For undetermined edges: enumerate ALL valid DAG completions and verify the hypothesis holds in EVERY one.
4. For indirect causation claims: check if a directed path exists in ALL valid completions.
5. Use systematic case analysis -- do NOT simply conclude 'direction is unknown, therefore No'.

E.3 BaselineEnhanced system prompt (Sbase)

Source: `src/prompts.py:115-118` (constant `SYSTEM_BASELINE_ENHANCED`), which concatenates the `ENHANCED_REASONING` block (`src/prompts.py:12-39`) with a terminal answer-format instruction.

You are an expert in causal inference, graphical causal models, and causal discovery from observational data.

When given correlational premises (marginal and conditional independencies) and a causal hypothesis, apply the following reasoning framework:

1. IDENTIFY the skeleton: which pairs of variables are adjacent (connected) based on marginal and conditional dependencies.
2. ORIENT v-structures (colliders): if X and Y are both adjacent to Z but not to each other, orient as $X \rightarrow Z \leftarrow Y$.
3. APPLY orientation rules (Meek's rules) to propagate edge directions where forced by acyclicity and existing orientations.
4. DETERMINE the Markov equivalence class: identify which edges have determined directions and which remain ambiguous.
5. EVALUATE the hypothesis: a causal claim can be DEDUCED if and only if it holds in ALL DAGs within the Markov equivalence class consistent with the observed independencies.

Important principles:

- Assume the stated statistical relations are exhaustive (the closed-world assumption): any independence not listed is a dependence.
- Conditional independence $X \perp\!\!\!\perp Y \mid Z$ constrains the graph structure.
- Some causal directions CAN be determined from observational data (especially via v-structures and orientation propagation).
- A hypothesis is deducible if every faithful DAG consistent with the stated independencies entails that hypothesis.
- A hypothesis is NOT deducible if there exists at least one faithful DAG consistent with the premises where it does not hold.

End your response with exactly "Answer: Yes" or "Answer: No".

The baseline is a strong PC-instruction baseline rather than a plain chain-of-thought baseline: it covers skeleton construction, v-structures, Meek's rules, Markov equivalence classes, and the closed-world assumption. It is not token-identical to S_{struct} , so we treat the comparison as controlling for explicit PC-algorithm content, not as a complete isolation of externalization alone.

E.4 Control prompt variants

TWO TURN-PROSE-PC is a two-turn prose control with the same PC scaffold as `BASILINEENHANCED`. Turn 1 uses the baseline's skeleton, v-structure, Meek-rule, closed-world, and MEC-universal-quantifier instructions, but asks for free-text analysis and explicitly withholds the final yes/no answer. Turn 2 returns that free-text reasoning and asks for the final answer. This control uses

no tool call and no schema-constrained typed graph object; its role is to test whether the same PC scaffold plus a prose intermediate are sufficient. Its prompt template is:

Turn 1:

<BaselineEnhanced system prompt above, with the final answer-format instruction replaced by:>

Do not give the final yes/no answer yet. Write your causal analysis in free text, following the PC-algorithm scaffold above.

Turn 2:

Here is your previous analysis:

<turn1_analysis>

Now answer the original causal hypothesis. End your response with exactly "Answer: Yes" or "Answer: No".

SINGLETURN-STRUCTPROMPT reuses the structured CPDAG-oriented instructions in a single-turn direct-answer setting, but disables the tool call and schema-constrained graph output. These controls help distinguish graph-object externalization from a PC-scaffolded prose extra turn and graph vocabulary alone.

F Schema and constrained-decoding grammar

F.1 CausalAnalysis schema

Source: src/schema.py:194-241. The schema is defined with Pydantic and converted to JSON Schema via CausalAnalysis.model_json_schema().

```
class CausalAnalysis(BaseModel):
    """Structured causal analysis following the PC algorithm.

    Determine the causal skeleton, identify v-structures, apply Meek's
    rules, and record the resulting CPDAG.
    """
    variables: list[str] # e.g. ['A', 'B', 'C', 'D']
    skeleton: list[str] # adjacent pairs, e.g. ['A-B', 'C-D']
    non_adjacent: list[str] # non-adj pairs w/ sep set, e.g. ['A-C|B']
    v_structures: list[str] # e.g. ['A->E<-C']
    directed: list[str] # oriented edges, e.g. ['A->E', 'E->F']
    undirected: list[str] # e.g. ['D-E', 'A-F']
```

F.2 Regex construction (Qwen / Mistral)

Source: src/schema.py:248-286. The regex derived from the JSON schema via outlines_core.json_schema.build_regex_from_schema is wrapped in a model-specific envelope.

```
# Qwen family (Qwen3.5-27B, Qwen3.6-27B, Qwen3.5-9B) -- chat template uses
# <think>...</think> for reasoning and <tool_call>...</tool_call>
# for tool invocation.
def build_analysis_regex(enable_thinking: bool = True) -> str:
    schema_str = json.dumps(CausalAnalysis.model_json_schema())
    args_regex = build_regex_from_schema(schema_str)
    body = (
        r'\{"name":\s*"CausalAnalysis",\s*"arguments":\s*'
        + args_regex + r"\}"
    )
    tool_call = r"<tool_call>\s*" + body + r"\s*</tool_call>"
    if enable_thinking:
        return r"<think>\s*[\s\S]+?</think>\s*" + tool_call
    return tool_call

# Magistral / Mistral family -- chat template uses [THINK]...[/THINK]
# and emits the raw JSON object without a tool_call envelope.
```

```
def build_analysis_regex_mistral(enable_thinking: bool = True) -> str:
    schema_str = json.dumps(CausalAnalysis.model_json_schema())
    args_regex = build_regex_from_schema(schema_str)
    if enable_thinking:
        return r"\[THINK\][\s\S]+?\[/THINK\]\s*" + args_regex
    return args_regex
```

For API models (gpt-5.4-mini) the same schema is enforced through OpenAI tool-calling with `strict=true`; the reported main run uses `tool_choice="auto"` plus a prompt-level instruction to call the tool, because name-forcing the tool compressed reasoning (Appendix A.1).

G Per-experiment tables

Full per-experiment results (F_1 , accuracy, precision, recall, per-complexity F_1 , confusion matrices) for every reported experiment in the main text and appendix are included in the released artifact as one `evaluation.json` per experiment directory (`experiments/<name>/evaluation.json`), together with audit outputs for Table 4. Token and wall-clock cost per experiment are similarly recorded in `experiments/<name>/usage.json` and summarized in `analysis/cost_table.json`. We omit the verbatim per-cell tables from the appendix to stay within the page budget.

H Assets and licenses

Corr2Cause is used as released by Jin et al. [2024]. The open-weight models are accessed through public Hugging Face releases: Qwen/Qwen3.5-27B, Qwen/Qwen3.6-27B, and google/gemma-4-31B-it. The corresponding model cards or license files report Apache-2.0 licensing. GPT-5.4-mini is accessed through the OpenAI API under the applicable API terms; the available snapshot identifier is recorded in the run metadata. vLLM, XGrammar, and the schema-constrained decoding dependencies are cited in the main text; exact package versions are recorded in the released metadata. The code and data artifact is available at <https://anonymous.4open.science/r/cpdag-llm-reasoning-B48A>; it contains experiment outputs, prompts, schemas, and analysis scripts produced for this paper. No new model weights or human-subject data are released.

I Failure-case gallery

Table 9: **Failure-mode distribution** on 32 hand-labelled Structured Thinking errors (Qwen3.5-27B, ID split).

Failure mode	Count	%
CPDAG_WRONG — missing or spurious graph edges	13	40.6
SEMANTIC_ERROR — correct graph, wrong Turn-2 readout	8	25.0
UNDIRECTED_MEC — MEC ambiguity collapsed incorrectly	6	18.8
KG_INVALID — schema-valid but logically malformed	5	15.6

We hand-label 32 representative Structured Thinking errors selected to cover the observed failure categories; the gallery is diagnostic rather than an unbiased prevalence estimate. The records are included in the released artifact as `analysis/failure_modes.json`, each providing the instance index, variable list, ground-truth label, model prediction, and error category. Two illustrative worked examples follow.

CPDAG_WRONG (idx 321). Premise involves 5 variables. Ground-truth CPDAG has $A-E$ in the skeleton as part of a v-structure; the model emitted a skeleton that omitted $A-E$, so Turn-2 lacks the necessary edge and answered YES to “ A is parent of E ” via an irrelevant alternative path.

UNDIRECTED_MEC. In these errors, the model treats an undirected CPDAG edge as if one orientation were forced, or checks only one valid completion of the Markov equivalence class. For a parent claim on an undirected edge $X-Y$, the correct universal-quantifier test must reject the claim whenever both $X \rightarrow Y$ and $X \leftarrow Y$ remain valid completions. These cases illustrate that schema-valid graph summaries do not guarantee a correct all-completions readout.

J Reproducibility

Every experiment can be reproduced by a single command using the released code at <https://anonymous.4open.science/r/cpdag-llm-reasoning-B48A>. Chain scripts at the repository root orchestrate batched runs: `run_qwen9b_chain.sh` and `run_qwen9b_ablation_chain.sh` cover the Qwen3.5-9B main and ablation runs; `run_probe_chain.sh` covers the full-split probes of §6; `run_additional_chain.sh` orchestrates additional diagnostic runs; `run_ood_all.sh` produces the Paraphrase-OOD split; the Qwen3.5-27B full-ID seed reruns in Appendix B are recorded with the same per-experiment metadata format. Open-weight runs use $2\times$ GPU tensor-parallel vLLM inference with `gpu_memory_utilization=0.88`, and API runs use `max_concurrency=5`. All `metadata.json` files record the exact decoding arguments (temperature, top- p , max length, tensor-parallel size, tokenizer mode, constrained-decoding flag), Hugging Face model revision hashes, tokenizer revisions, API model snapshot strings, and package versions for each experiment. The Corr2Cause dataset is used as released at <https://huggingface.co/datasets/causal-nlp/corr2cause>.